

OOP

programare orientată pe obiecte

CURS11: METRICS FOR OBJECT ORIENTED DESIGN

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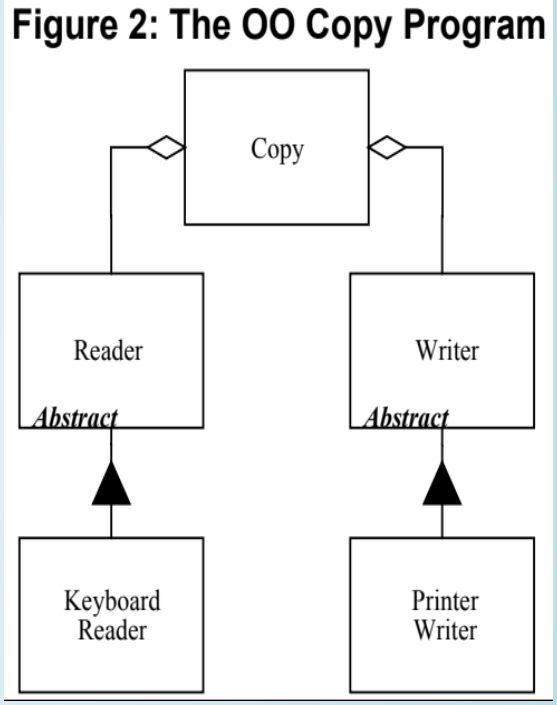
Cum ne organizam codul in asa fel incat...

- sa fie usor de citit
- sa fie usor de testat (automat)
- sa fie usor de extins
- sa fie usor sa impartim lucrul cu colegii

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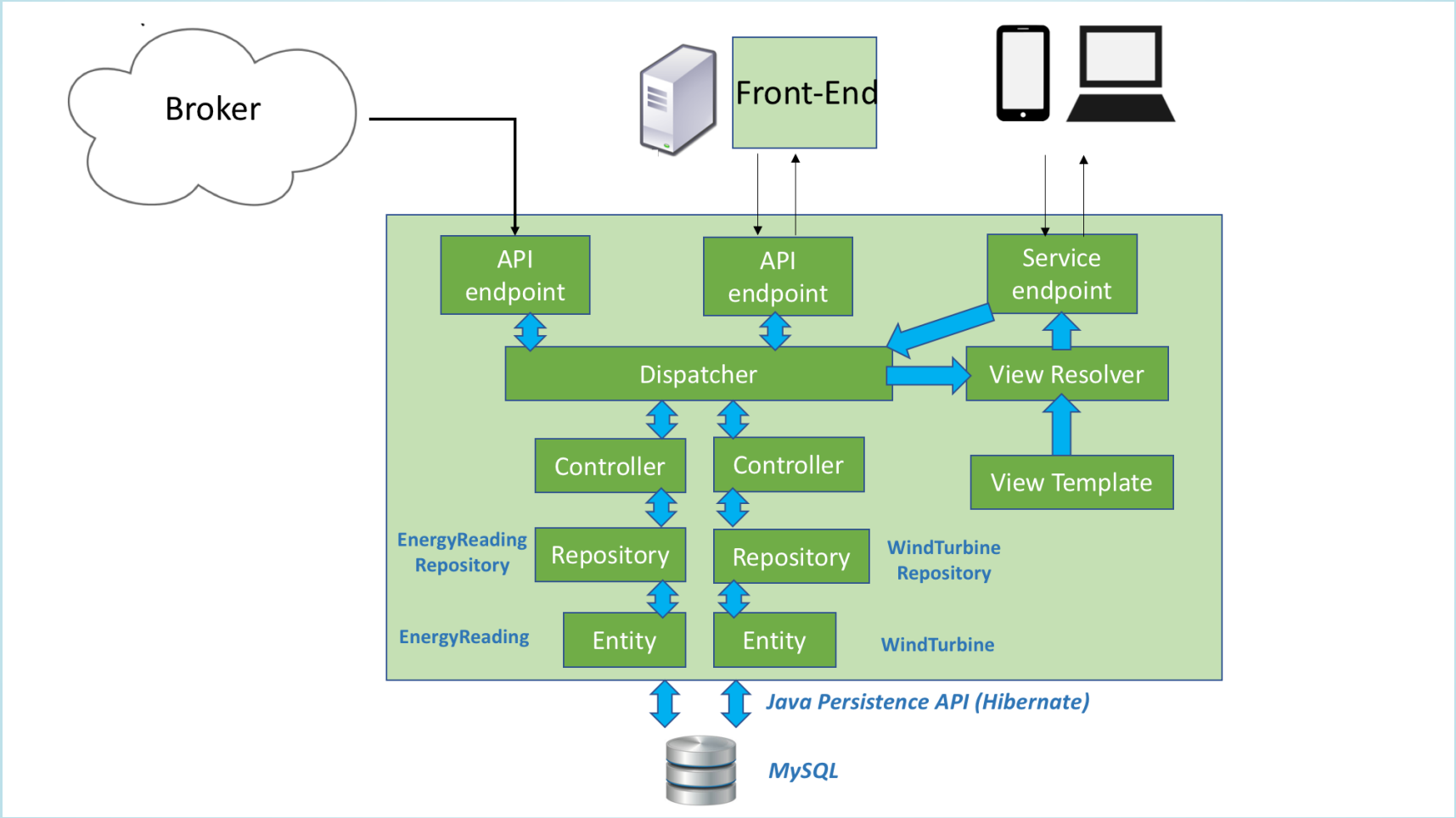
Object Oriented Design Principles (SOLID)

- SRP: Single Responsibility Principle
- OCP: Open-Closed Principle
- LSP: Liskov Substitution Principle
- ISP: Interface Segregation Principle
- DIP: Dependency Inversion Principle



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Frameworks (e.g. Spring for Java)



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Chidamber, S.R. and Kemerer, C.F. 1994. “A metrics suite for object oriented design,” IEEE Transactions on Software Engineering

Metric	Object Definition	Object Semantics	Relationships between Objects
WMC	✔	✔	
DIT	✔		
NOC	✔		
RFC		✔	✔
CBO			✔
LCOM		✔	

Table 1: Mapping of Metrics to Booch OOD Steps

- Weighted Methods per Class (WMC)
- Depth of Inheritance Tree (DIT)
- Number of Children (NOC)
- Coupling between Objects (CBO)
- Response For a Class (RFC)
- Lack of Cohesion in Methods (LCOM)

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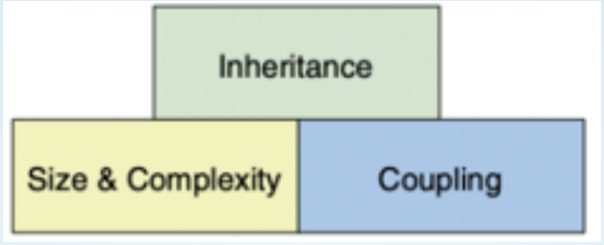
"The suite provides senior designers and managers with an indication of the integrity of the design... "

"... identify areas of the application that might require more rigurous testing... "

"... added insights gained about trade-offs made by the designers between conflicting requirements such as increased reuse (via more Inheritance) and ease of testing (via a less complex Inheritance hierarchy)"

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Lanza, M., and R. Marinescu. "Object-Oriented Metrics in Practice. 2005." Google Scholar Digital Library.



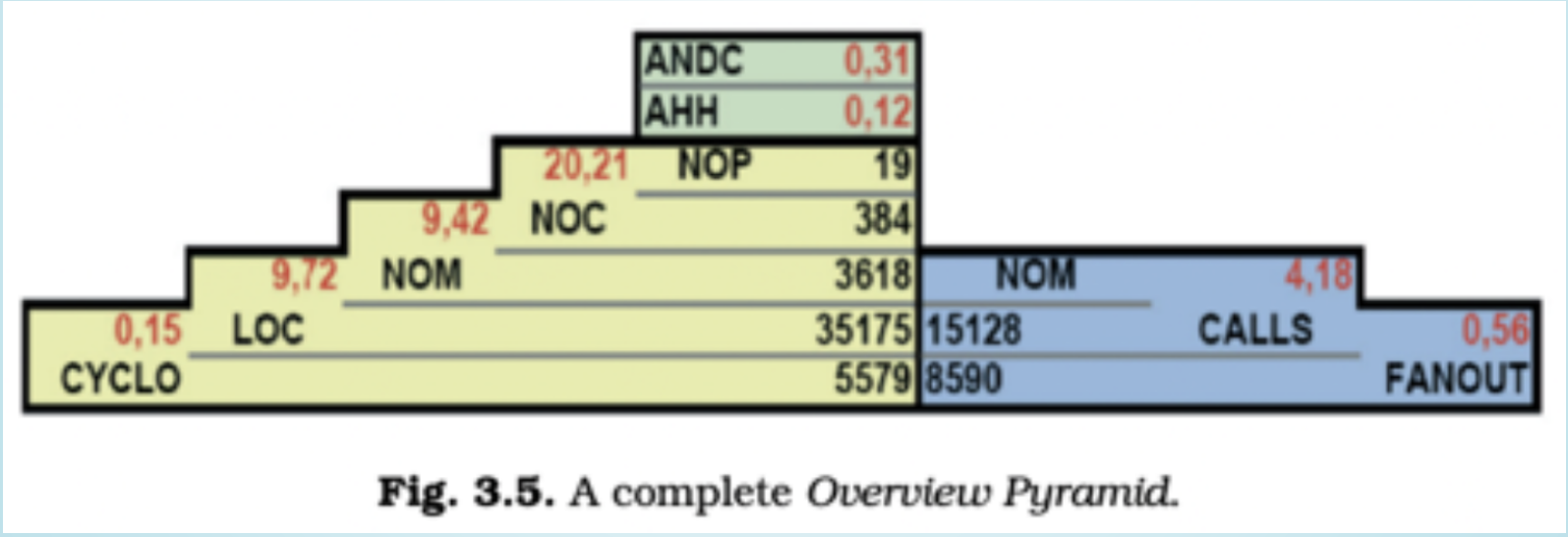


Fig. 3.5. A complete *Overview Pyramid.*

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SonarQube: Rules > Language: Java

- Methods should not have too many lines
- Inheritance tree of classes should not be too deep
- Files should not have too many lines of code
- Classes should not be coupled to too many other classes (Single Responsibility Principle)
- Methods should not be too complex (cyclomatic complexity)
- Cognitive Complexity of methods should not be too high

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Cyclomatic Complexity measures the number of linearly-independent paths through a program module

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In general, formula McCabe aplicata asupra control-flow graph:
M = E - N + 2P

Cyclomatic Complexity on Wikipedia

In SonarQube, "Keywords incrementing the complexity: if, for, while, case, catch, throw, &&, ||, ?"

How does Sonar calculate the cyclomatic complexity?

Cognitive Complexity is a measure of how hard the control flow of a method is to understand (a programmer's cognitive effort required to understand those flows)

White paper from Sonar on Cognitive Complexity

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Cyclomatic Complexity: increment breaks in the control-flow

```
int sumOfPrimes(int max) { // +1
  int total = 0;
  OUT: for (int i = 1; i <= max; ++i) { // +1
    for (int j = 2; j < i; ++j) { // +1
      if (i % j == 0) { // +1
        continue OUT;
      }
    }
    total += i;
  }
  return total;
} // Cyclomatic Complexity 4
```

```
String getWords(int number) { // +1
  switch (number) {
    case 1: // +1
      return "one";
    case 2: // +1
      return "a couple";
    case 3: // +1
      return "a few";
    default:
      return "lots";
  }
} // Cyclomatic Complexity 4
```

Metrics for OOD

ale

Cognitive Complexity: increment breaks in the control-flow, increment nested structures, ignore shorthand

```
int sumOfPrimes(int max) {
  int total = 0;
  OUT: for (int i = 1; i <= max; ++i) { // +1
    for (int j = 2; j < i; ++j) { // +2
      if (i % j == 0) { // +3
        continue OUT;  // +1
      }
    }
    total += i;
  }
  return total;
} // Cognitive Complexity 7
```

```
String getWords(int number) {
  switch (number) {   // +1
    case 1:
      return "one";
    case 2:
      return "a couple";
    case 3:
      return "a few";
    default:
      return "lots";
```

```
  } // Cyclomatic Complexity 1
}
```

Metrics for OOD

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The processes of writing and maintaining code are human processes. Their outputs must adhere to mathematical models, but they do not fit into mathematical models themselves. This is why mathematical models are inadequate to assess the effort they require.

White paper from Sonar on Cognitive Complexity